Synopsis of API 610, 12th Edition (Major changes wrt 11th Edition)

API 610, the standard specifies requirements for Centrifugal Pumps, including pumps running in reverse as hydraulic power recovery turbines (HPRTs), for use in petroleum, petrochemical, and gas industry process services. Latest 12th Edition has just been released in the last week of January, 2021.

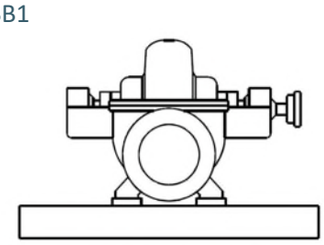
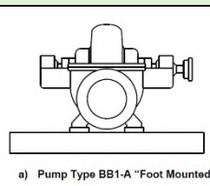
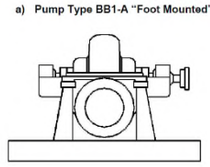
Key features and major changes of 12th edition, are highlighted in this write up with the objective to support Pump professionals with quick overview who already know about the API 610, 11th edition, which is 10 years old. Deliberately, I have avoided to discuss on interpretation, implications and effect of these changes to OEM or Purchaser. This is not a complete highlight of 12th edition. You may not find many fundamental and important recommendations in this article, if those are the same as 11th edition. Reference of API section is provided in parenthesis. Off course, this list is selective, not exhaustive.

- **Basic Overview**

Section	API 610 11th Edition	API 610 12th Edition
TOC	Total 10 Main sections + Annexures from A to O	Total 10 Main Sections + Annexures from A to O
Additional annexure	Annex O: API regional annexure	Figure N3: Data list, provide summary of all the field of API datasheet with page number reference
		Annex O: (informative) Special-purpose Centrifugal Pumps.
Released in	September 2010	January 2021
Total Pages	205	223

Acronyms and Abbreviations are provided with around 60 items (*Ref. 3.2*)

- **Pump Classification (*Ref. 4.2, Table 1*)**

API 610 11th Edition	API 610 12th Edition
One or Two stage, Axial split between bearing pump is called BB1  Figure 7 — Pump type BB1	BB1 is with two variants. Foot mounted → BB1-A Near centerline mounted → BB1-B  a) Pump Type BB1-A "Foot Mounted" For BB2 and BB3, Requirement of Centerline supported is added.  b) Pump Type BB1-B "Near-centerline Mounted"

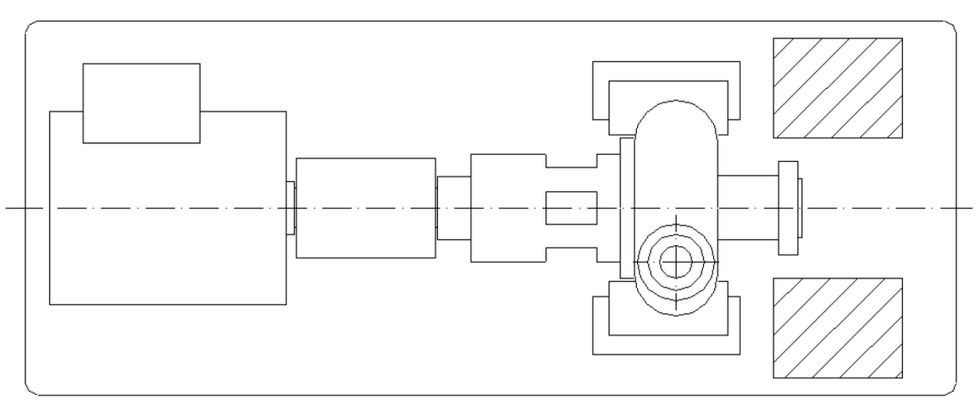
- **Basic design and selection *(Ref. 6.1)***

  No major changes in the 12th edition as far as core design features of pump is concerned. Additional requirement focusing on improved equipment reliability are addressed below. Requirement of field proven record and API 691 is introduced

Section	API 610, 11th edition	API 610, 12th edition, different or additional requirement
Reliability concept	Shall be designed and constructed of 20 years and at least 3 years of uninterrupted operation.	Only equipment that is field proven, as defined by the Purchaser, is acceptable, API 691 can provide guidance on this. In the event no such equipment is available, the vendor shall submit an explanation of how their proposed equipment can be considered field proven.
		The vendor shall advise in the proposal any component designed for a finite life.
		The purchaser shall specify if equipment will be supplied in accordance with API 691.
Parallel Operation		Additional recommendations: the head values of the pumps at any given flow within the preferred operating range shall be within 3 % of each other for pumps larger than 3 in. (80 mm) discharge.
Curve Shape		Pumps with a continuously rising head curve are preferred for all applications, but this is not possible with all pump types. Head curve shape is dependent on several factors specific to the pumps hydraulic design.
End of curve	Not available	The "end of curve flow" is defined as 120 % of the BEP flowrate.
Viscosity correction	performance corrected in accordance with ISO/TR 17766	performance corrected in accordance with HI 9.6.7. Both the standards are equivalent.

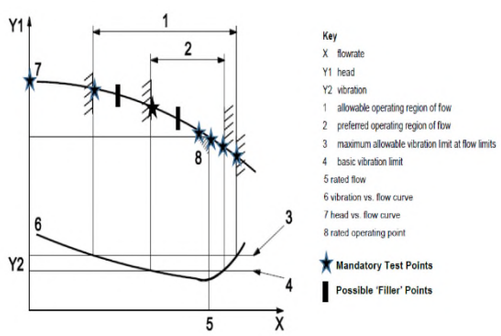
Synopsis of API 610, 12th Edition (Major changes wrt 11th Edition)

- **Accessories (*Ref.7)***

Accessories	Changes or additional points in 12th edition
Coupling	Flexible elements shall be non-lubricated metal type of corrosion-resistant material.
	If specified, major coupling components shall be balanced in accordance with ISO 21940-11, to the balance grade specified by the purchaser. According to 11th edition it is ISO 1940-11, grade G6.3
Guard	Separate sub-section is available on Guard.
Coupling Guard	Unless otherwise specified, guards between drivers and driven equipment and between the bearing housing and seal gland shall be supplied and mounted by the vendor with unit responsibility.
Shaft Guard (New concept)	Exposed shaft areas including the area between pump bearing housing(s) and mechanical seal(s) shall have a shaft guard. The guard shall meet the following requirements: a) prevent personnel from contacting moving parts during operation of the pump; allowable opening dimensions shall comply with specified standards, such as EN 953 or ISO 14120; b) sufficiently vented to prevent the accumulation of seal emissions, liquid, or vapor; c) allow visual inspection of the seal without removal of guard; d) constructed of steel, stainless steel, brass, or aluminum materials, as suitable; e) fabricated from sheet (solid or perforated), plate, expanded metal, or woven wire and securely fastened to the pump.
Base Plate	Single-piece baseplates designed for grouting shall be furnished for horizontal pumps. The purchaser shall specify the type and options as follows: a) Flat deck plate with a sloped gutter drain, b) Sloped full deck plate c) Sloped partial deck plate, d) Open deck version of the above with no deck/top plate e) Non-grouted baseplate of one of the versions above where the baseplate and pedestal support assembly shall be sufficiently rigid to be mounted without a grout fill, f) Non-grouted baseplate as in Item e) with a gimbal mount, three-point mount, anti-vibration mount (AVM) spring mount, or other type of mount.
	Baseplates for OH2 pumps shall have nothing (auxiliaries or seal flush plan) mounted beside or above the coupling or bearing housing. If the seal flush plan and/or auxiliaries are specified to be mounted on the baseplate, the increased length standard baseplate shall be used and the auxiliaries and/or seal flush plan shall be mounted adjacent to the suction nozzle (see Figure 40). 
	Annexure D : Base plate numbers are 2.5 to 12 whereas 11th edition have 0.5 to 12.
Vertical Pump Driver	12th edition: Shaft-to-driver mating face perpendicularity and surface flatness. 0.002 in./ft (0.17 mm/m) 11th edition: Shaft-to-driver mating face perpendicularity and surface flatness. 25 µm TIR
Piping and Appurtenances	12th edition: Piping shall be in accordance with API 614, API 682, and this standard. API 682 and this standard take precedence in case of conflicts with API 614. 11th edition: Piping shall be in accordance with ISO 10438.
Instrumentation	12th edition: Accordance with API 614 11th edition: Accordance with ISO 10438

Synopsis of API 610, 12th Edition (Major changes wrt 11th Edition)

- **Pump Performance standards (*Ref.8.3*)**

Section	API 610, 11th edition	API 610, 12th edition, difference or additions
Performance and NPSH test standard	ISO 9906, Grade 1. ANSI/HI 1.6 - Centrifugal pump, ANSI/HI 2.6 for vertical pump	HI 14.6 (ISO 9906), Grade 1
Performance test points	1. Shutoff (No vibration data) 2. Minimum continuous stable flow 3. Midway between minimum and rated flow 4. Between 95% and 99% of rated flow 5. Between rated flow and 105% of rated flow 6. Maximum allowable flow (end of allowable operating region)	1. Shutoff (no vibration data required) 2. Minimum continuous stable flow 3. Approx. halfway b/w minimum continuous stable flow and minimum preferred operating flow 4. Minimum preferred operating flow 5. Approx. halfway b/w minimum preferred operating flow and rated flow 6. B/w 95 % and 99 % of rated flow 7. B/w rated flow and 105 % of rated flow 8. End of preferred operating region 9. End of allowable operating region if different from the end of the preferred operating region.  For units with BEP less than 11 m3/h, Point 3) and Point 5) are not required.
Explanation on why not to test certain pump at shutoff		Higher energy, integrally geared, or multistage pumps exhibit a high rate of increase of the temperature at shutoff that makes it not feasible and/or not safe to test them at shut off. The temperature increase is strictly related to the power density. The power density, PD, can be approximated as: $$PD = P_{rated}/D_{imp}^2 \, X \, D_{nozzle}$$ Where, P_{rated} → Rated power per stage in hp (or MW) with water; D_{imp} → Rated impeller diameter in in. (or m); D_{nozzle} → Nominal outlet flange diameter in in. (or m). For double-suction, single-stage pumps, it is the inlet flange diameter. Typical critical value of PD, beyond which it is suggested that the pump not be operated at shutoff during performance test is 0.286 hp/in.3 (13 MW/m3).
Site performance recommendations (*Ref 6.1.34*)		Many factors can adversely affect site performance. These factors include such items as piping loads, alignment at operating conditions, supporting structure, handling during shipment, and handling and assembly at the site. If specified, the vendor's representative shall witness: a) a check of the piping alignment performed by unfastening the major flanged connections of the equipment; b) the initial shaft alignment check at ambient conditions; c) shaft alignment at operating temperature, i.e. hot alignment check

- **Material of Construction (*Annexure G and H*)**

  - ✓ Material class I-1, I-2, S-1, and S-3 are completely removed from the selection in annexure H. Minimum material class is now S4. Few minor changes in the material class S.
  - ✓ List of services and operating temperature are also changed in annexure G.
  - ✓ Material class selection for few services is changed as per the annexure G.

Synopsis of API 610, 12th Edition (Major changes wrt 11th Edition)

- **API Data sheet (*Annexure. N*)**

  - ✓ API data sheet in the 12th edition is almost the same as 11th edition, except minor lay out and cosmetic changes.
  - ✓ Data list is provided after the datasheet annexure, covers all the terms used in the datasheet with its reference page number on this API standard to get additional clarification.
  - ✓ Format of project data sheet is provided to record and communicate on site design data and utility condition.

- **Contract documents and Engineering data (*Annexure. I)***

  - ✓ Vendors data requirement such as contract data, proposal, Drawings and technical data, recommended spares data, manuals etc. is part of the section 10 of API 610, 11th edition. whereas in 12th edition it is covered under the annexure L, Contract documents and Engineering design data.

- **Special purpose Centrifugal pump (*Annexure. O)***

  - ✓ There is an exclusive annexure with topic on special-purpose centrifugal pump**.** It is one that meets one or more of the following criteria:
    a) Operating in a severe, critical application;
    b) A prototype—a design that is sufficiently different from existing designs to warrant, in either the manufacturer's or the purchaser's opinion, the extent of engineering analysis and manufacturing process specified in this Annex O;
    c) High energy levels.
  - ✓ Special-purpose pumps can have large shafts at high speed or can have other duties that are beyond normal mechanical seal operating parameters. For such pumps, the parties are to consider the following steps in seal selection and especially prototype testing:
    a) Abide by requirements of API 610 and API 682 where applicable,
    b) Consider dry gas seals (DGSs) for light hydrocarbons below 0.70 specific gravity (as referenced on Table 14) and CO2,
    c) Consider API 682 qualification test of prototype seals to establish seal performance before releasing for manufacture.
